

Processando config 1, seed 1, teste: Rabbit
XX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

smultin_MultinomialBitsOver test:

N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = $1 / 32768$
EColl = $n^2 / (2k)$ = 128
Hashing = TRUE

Collision test
CollisionOver: density = $n / k = 1 / 32768.00$
Expected number of collisions = $\mu = 128.00$

Results of CollisionOver test:

POISSON approximation :
Expected number of collisions = $N \cdot \mu$: 256.00
Observed number of collisions : 271
p-value of test : 0.18

Total number of cells containing j balls

j = 0	:	549739036943
j = 1	:	16776674
j = 2	:	271
j = 3	:	0
j = 4	:	0
j = 5	:	0

CPU time used : 00:00:04.53

Generator state:
16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 18.00
p-value of test : 0.12

CPU time used : 00:00:00.04

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 11.00
p-value of test : 3.1e-5 *****

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 2.39
p-value of test : 8.3e-3

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 12.50
p-value of test : 0.09

Normal statistic for number of jumps : -51.45
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.06

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.32
p-value of test : 0.38

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.85
p-value of test : 0.80

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 3.42e-3
p-value of test : 0.99

Kolmogorov-Smirnov- statistic = D- : 0.056
p-value of test : 0.04

Anderson-Darling statistic = A2 : 2.76
p-value of test : 0.04

CPU time used : 00:00:00.11

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 5.62
p-value of test : 0.13

Global longest run of 1 : 25.00
p-value of test : 0.22

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 70.25
p-value of test : 0.97

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 22.01
p-value of test : 0.46

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.23
p-value of test : 0.41

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -1.96
p-value of test : 0.97

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.58
p-value of test : 0.28

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 162.48
p-value of test : 0.78

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 300.79
p-value of test : 0.56

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 2333.37
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :    1.01
p-value of test          :    0.16
-----
```

```
CPU time used            : 00:00:00.02
```

```
Generator state:
269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :   -1.56
p-value of test          :    0.94
-----
```

```
CPU time used            : 00:00:00.00
```

```
Generator state:
286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin
```

```
sstring_Run test:
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```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355447
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 38.12
p-value of test : 0.37
-----
```

```
-----
Total number of bits: 13417376
```

```
Normal statistic for number of bits : -0.85
p-value of test : 0.80
```

CPU time used : 00:00:00.04

Generator state:
299694464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 267.05
p-value of test : eps *****

CPU time used : 00:00:00.07

Generator state:
316471680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333162880 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42	
ChiSquare statistic	:	1078.76	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	42	
ChiSquare statistic	:	452.02	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	64	
ChiSquare statistic	:	98.82	
p-value of test	:	3.4e-3	

Test on the values of the Statistic R

Number of degrees of freedom	:	36	
ChiSquare statistic	:	21.92	
p-value of test	:	0.97	

Test on the values of the Statistic C

Number of degrees of freedom	:	22	
ChiSquare statistic	:	13.33	
p-value of test	:	0.92	

CPU time used : 00:00:00.04

Generator state:
349940096 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94	
ChiSquare statistic	:	459.48	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	94	
ChiSquare statistic	:	486.98	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	512	
ChiSquare statistic	:	605.10	
p-value of test	:	2.8e-3	

Test on the values of the Statistic R

Number of degrees of freedom	:	90	
ChiSquare statistic	:	93.48	
p-value of test	:	0.38	

Test on the values of the Statistic C

Number of degrees of freedom	:	50	
ChiSquare statistic	:	52.64	
p-value of test	:	0.37	

CPU time used : 00:00:00.03

Generator state:

366717312 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	132.20
p-value of test	:	0.43

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	170.90
p-value of test	:	0.01

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	169.27
p-value of test	:	0.31

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	121.47
p-value of test	:	0.69

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	79.13
p-value of test	:	0.76

CPU time used : 00:00:00.03

Generator state:

383494112 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.00
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

	Test	p-value
3	ClosePairsBitMatch, t = 4	3.1e-5
5	LinearComp	1 - eps1
17	HammingIndep, L = 64	eps
21	MatrixRank, 32 x 32	eps
22	MatrixRank, 320 x 320	eps
24	RandomWalk1 H	eps
24	RandomWalk1 M	eps
25	RandomWalk1 H (L = 1024)	eps
25	RandomWalk1 M (L = 1024)	eps

All other tests were passed